Yushi Zeng

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EDUCATION

South China University of Technology — Guangzhou, China
Doctor of Engineering in Future Technology (Advisor: Dr. Yi Cai) — Sept. 2023 - Present
South China University of Technology — Guangzhou, China
Master of Engineering in Software Engineering (Advisor: Dr. Yi Cai) — Sept. 2020 - Jun. 2023
Jishou University — Hunan, China
Bachelor of Engineering in Software Engineering — GPA:3.47/4.00 — Sept.2016-Jun.2020

RESEARCH INTERESTS

Sentiment Analysis, Knowledge Graph, Cloud-Device Collaborative Learning, Multimodal Learning.

PUBLICATIONS

[1] **Zeng, Y.**, Ren, H., Cai, Y., Li, Y., Qin, J., & Wang, Y. (2026). Modality augmentation and task-aware dual-modal LoRAs for multi-task multimodal federated learning. *Information Processing & Management*, 63(4), 104601. **(IF 8.3, JCR Q1)**

[2] Li, Y. (Supervisor), **Zeng, Y.**, Jiang, D., Wang, Y., & Lu, G. (2026). Rethinking the knowledge gap between cloud and device models for effective co-adaptation. *IEEE Transactions on Circuits and Systems for Video Technology*, 36(7), 9888–9900. **(IF 10.8, JCR Q1, Co-first author)**

[3] **Zeng, Y.**, Ren, H., Cai, Y., Li, Y., Wang, Y., & Li, Q. (2026). Class-incremental cloud-device collaborative adaptation with contrastive learning in dynamic changing environments. *IEEE Transactions on Neural Networks and Learning Systems*, 37(8), 3970–3984. **(IF 9.7, JCR Q1)**

[4] **Zeng, Y.**, Wang, G., Ren, H., Cai, Y., Leung, H. F., Li, Q., & Huang, Q. (2023). A knowledge-enhanced and topic-guided domain adaptation model for aspect-based sentiment analysis. *IEEE Transactions on Affective Computing*, 15(2), 709–721. **(IF 11.7, JCR Q1)**

[5] **Zeng, Y.**, Wang, G., Ren, H., & Cai, Y. (2022, June). Enhance cross-domain aspect-based sentiment analysis by incorporating commonsense relational structure (Student Abstract). In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 36, No. 11, pp. 13105–13106). **(CCF-A)**

[6] Ren, H., **Zeng, Y.**, Cai, Y., Ye, Z., Yuan, L., & Zhu, P. (2024, May). Grounded multimodal procedural entity recognition for procedural documents: A new dataset and baseline. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)* (pp. 7971–7981). **(CCF-B; Co-first author)**

[7] Ren, H., **Zeng, Y.**, Cai, Y., Zhou, B., & Lian, Z. (2023, July). Constructing procedural graphs with multiple dependency relations: A new dataset and baseline. In *Findings of the Association for Computational Linguistics: ACL 2023* (pp. 8474–8486). **(CCF-A; Co-first author)**

[8] Zhou, B., Ren, H., Cai, Y., Lian, Z., Zhu, P., **Zeng, Y.**. MPPQA: Structure-Aware Extractive Multi-span Question Answering for Procedural Documents. *Natural Language Processing and Chinese Computing. (NLPCC 2024.)*, vol 15359. **(CCF-B)**

[9] H. Ren, Y. Cai, **Y. Zeng**, J. Ye, H. -f. Leung and Q. Li, (2022) "Aspect-Opinion Correlation Aware and Knowledge-Expansion Few Shot Cross-Domain Sentiment Classification," in *IEEE Transactions on Affective Computing*. vol. 13, no. 4, pp. 1691-1704 **(IF 11.7, JCR Q1)**

[10] Ren, H., Cai, Y., & **Zeng, Y.**. (2022). Aspect-Opinion Sentiment Alignment for Cross-Domain Sentiment Analysis (Student Abstract). *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(11), **(CCF-A)**

RESEARCH EXPERIENCE

Research on Standard knowledge graph construction and application — Jingshan WPS
Research Member — Sept 2020 — Nov 2021

- Knowledge Graph Construction: The Schema construction for the standard domain and the knowledge extraction Extract the entity-relation triples by utilizing the inherent document structure features of the standard documents. Moreover, the BILSTM-CRF and BERT-CRF models are used to extract the fine-grained entities. Then, the extracted standard knowledge triples are stored in the neo4j database and we conduct the visualization analysis.
- tandard Knowledge Recommendation: We adopt a content-based recommendation method which uses the user-entered keywords to retrieve and link the relevant entities from standard knowledge graph.
- Standard conflict detection: We recognize the candidate entity from the standard drafts and kink them from the knowledge graph and identify the suspected conflicting entries through the text comparison.

INTERNSHIP EXPERIENCE

Huawei | "AI Applications in Wireless Information Experience" — Shenzhen, China
IT Platform Department 3 — Jun. 2022 — Sep. 2022

- Worked on an internal Huawei project involving heterogeneous knowledge bases, solving cross-graph retrieval challenges by linking

product and terminology knowledge graphs to improve the relevance of technical entity retrieval.
- Built an entity recognition, linking, and disambiguation pipeline with Jieba, pre-trained BERT embeddings, and cosine similarity, reducing ambiguity in product names and attribute descriptions and improving matching accuracy.
- Improved system deployment and API performance by designing a multi-threaded access strategy, raising entity linking accuracy to over 90% and ensuring 99% of response times remained under 200 ms.

TEACHING EXPERIENCE

South China University of Technology Guangzhou, China
Graduate Teaching Assistant, Computer Architecture [Fall 2023]

- Graded homework assignments, held regular office hours for student Q&A.
- Assisted the instructor in supervising and guiding lab sessions for computer architecture experiments.

AWARDS & COMPETITION

- Outstanding Student, 2025 Tencent Rhino-Bird Open Source Talent Development Program 2025
- Wuliangye Outstanding Student Scholarship 2023
- China National Scholarship 2022
- First-Class Postgraduate Scholarship 2022
- AAAI Student Scholarship Program 2022
- Certificate of Completion, Summer Research Experience Programme for Mainland and Taiwan Students, The Chinese University of Hong Kong 2022
- Outstanding Graduate, Jishou University 2020
- First-Class and Second-Class Scholarships, Jishou University (Awarded over Four Consecutive Years) 2017–2020
- Third Prize, The 10th Lanqiao Cup National Software and Information Technology Competition 2019

SKILLS & INTERESTS

- **Languages:** Mandarin (Native), Cantonese (Fluent), English (Fluent,CET6)
- **Programming Languages:** Python (PyTorch), JAVA, C
- **Software and Tools:** Microsoft Office (Word, Excel, and PowerPoint), Zotero, LaTex, Markdown
- **Interests:** Badminton, Fitness